

Disproof of Conjecture 00000001068

The maximal sum-free subsets of $\mathbb{F}_p$ are *not* exactly the intervals $((p+1)/3, 2(p-1)/3)$ up to dilation

TLMC submission

October 2, 2026

1 The conjecture

Conjecture 00000001068 (as stated). *Definition: a sum-free set is a set A with $A + A$ disjoint from A . The maximal sum-free subsets of $\mathbb{F}_p$ are exactly the intervals $((p+1)/3, 2(p-1)/3)$, uniquely up to dilation (the complete Diananda–Yap characterization over finite fields).*

2 Counterexample: $p = 11$

Let $A = \{4, 5, 6, 7\} \subseteq \mathbb{F}_{11}$.

Theorem. *A is an inclusion-maximal sum-free subset of $\mathbb{F}_{11}$, but A is not a dilation of $((p+1)/3, 2(p-1)/3) = (4, 20/3)$. Hence the conjecture is false.*

Proof. Sum-free. Computing all pairwise sums in $\mathbb{F}_{11}$:

$$A + A = \{8, 9, 10, 0, 9, 10, 0, 1, 10, 0, 1, 2, 0, 1, 2, 3\} = \{0, 1, 2, 3, 8, 9, 10\} = \mathbb{F}_{11} \setminus A,$$

which is disjoint from A . (So A is even a *maximum* sum-free set: $|A| = 4$, and no sum-free set in $\mathbb{F}_{11}$ contains a proper superset of A .)

Inclusion-maximal. Every $x \in \mathbb{F}_{11} \setminus A$ lies in $A + A$; hence $x \in (A \cup \{x\}) + (A \cup \{x\})$ and $A \cup \{x\}$ is not sum-free. So A is an inclusion-maximal sum-free subset of $\mathbb{F}_{11}$.

The conjectured family. With rational (open) interval endpoints,

$$((p+1)/3, 2(p-1)/3) = (4, 20/3) \cap \mathbb{Z} = \{5, 6\} \subseteq \mathbb{F}_{11},$$

which has only $|I| = 2$ elements. Every dilation cI with $c \in \mathbb{F}_{11}^\times$ again has exactly 2 elements, and $0 \cdot I = \{0\}$. Exhaustively, over all $c \in \{0, 1, \dots, 10\}$ one has $c \cdot \{5, 6\} \neq \{4, 5, 6, 7\}$ (the distinct images are $\{0\}, \{5, 6\}, \{10, 1\}, \{4, 7\}, \{9, 2\}, \{3, 8\}$).

Since $|A| = 4 \neq 2$, A is an inclusion-maximal sum-free subset of $\mathbb{F}_{11}$ that is not of the form $c((p+1)/3, 2(p-1)/3)$. The claimed characterization fails. $\square$

3 Remarks

The failure is the open-interval formula itself. The classical Diananda–Yap extremal family in $\mathbb{Z}_p$ is the *closed* integer interval $\{\lfloor p/3 \rfloor + 1, \dots, \lfloor 2p/3 \rfloor\}$, which at $p = 11$ is $\{4, 5, 6, 7\} = A$ itself (size $\lfloor 2p/3 \rfloor - \lfloor p/3 \rfloor = 4$); the conjecture's open interval $(4, 20/3)$ drops the two endpoints and keeps only 2 points, so the family it defines is not maximal.

4 Machine verification

- `reproduce.py` recomputes $A + A$, the interval, all dilations, sum-freeness, and inclusion-maximality from scratch; all checks pass.
- `lean4/Main.lean` formalizes the counterexample in Lean 4 (v4.33.1, core only): `A_sum_free`, `A_maximal`, `A_len`, and `A_not_dilation` combine into `counterexample_1068`. All proofs are `rfl` over explicit $\mathbb{F}_{11}$ computations. The audit in `Check.lean` reports **7/7 theorems depend on no axioms** (no `sorry`, no `propext`, no `Classical.choice`, no `Quot.sound`).